Language Technology

Chapter 12: Words, Parts of Speech, and Morphology

https://link.springer.com/chapter/10.1007/978-3-031-57549-5_12

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September 10, 2026



The Parts of Speech

Parts of speech (POS) are classes that correspond to lexical (or word) categories.
Plato distinguished between the verb and the noun.
After him, the inventory of word categories continued to evolve and expand until Dionysius Thrax formalized and stabilized it.
Aelius Donatus later popularized the list of the eight classical parts of speech: noun, pronoun, verb, participle, conjunction, adverb, preposition, and interjection.
Grammarians have adopted these categories for most European languages, even though the classification remains somewhat arbitrary.
Parts of speech fall into two main classes: the open class and the closed class.



Parts of Speech: Open-Class Words

POS	English	French	German
Nouns	*name, Frank*	*nom, François*	*Name, Franz*
Adjectives	*big, good*	*grand, bon*	*groß, gut*
Verbs	*to swim*	*nager*	*schwimmen*
Adverbs	*rather, very, only*	*plutôt, très, uniquement*	*fast, nur, sehr, endlich*



Parts of Speech: Closed-Class Words

POS	English	French	German
Determiners	*the, several, my*	*le, plusieurs, mon*	*der, mehrere, mein*
Pronouns	*he, she, it*	*il, elle, lui*	*er, sie, ihm*
Prepositions	*to, of*	*vers, de*	*nach, von*
Conjunctions	*and, or*	*et, ou*	*und, oder*
Auxiliaries and modals	*be, have, will, would*	*être, avoir, pouvoir*	*sein, haben, können*



Annotation with Parts of Speech

Sentence:

That round table might collapse

Annotation:

Word	**Part of speech**	**POS tag**
that	Determiner	DET
round	Adjective	ADJ
table	Noun	NOUN
might	Modal verb	AUX
collapse	Verb	VERB

Automatic annotation relies on predefined POS tagsets, such as the Penn Treebank tagset for English.



Training Sets: The CoNLL Format

The CoNLL format is a tabular format used to distribute annotated texts.
It was originally created for the shared tasks of the Conference on Natural Language Learning.
The precise annotation scheme has varied over the years; we use CoNLL-U, the most recent version.
Annotation of the Spanish sentence:

> *La reestructuración de los otros bancos checos se está acompañando por la reducción del personal*
> *'The restructuring of the other Czech banks is being accompanied by a reduction in staff'*



Example of Annotation (CoNLL-U)

La reestructuración de los otros bancos checos se está acompañando por la reducción del personal

ID	FORM	LEMMA	UPOS	FEATS
1	La	el	DET	Definite=Def|Gender=Fem|Number=Sing|PronType=Art
2	reestructuración	reestructuración	NOUN	Gender=Fem|Number=Sing
3	de	de	ADP	AdpType=Prep
4	los	el	DET	Definite=Def|Gender=Masc|Number=Plur|PronType=Art
5	otros	otro	DET	Gender=Masc|Number=Plur|PronType=Ind
6	bancos	banco	NOUN	Gender=Masc|Number=Plur
7	checos	checo	ADJ	Gender=Masc|Number=Plur
8	se	se	PRON	Case=Acc|Person=3|PrepCase=Npr|PronType=Prs|Reflex=Yes
9	está	estar	AUX	Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin
10	acompañando	acompañar	VERB	VerbForm=Ger
11	por	por	ADP	AdpType=Prep
12	la	el	DET	Definite=Def|Gender=Fem|Number=Sing|PronType=Art
13	reducción	reducción	NOUN	Gender=Fem|Number=Sing
14	del	del	ADP	AdpType=Preppron
15	personal	personal	NOUN	Gender=Masc|Number=Sing
16	.	.	PUNCT	PunctType=Peri



Part-of-Speech Annotation (CoNLL 2000)

Annotation of: *He reckons the current account deficit will narrow to only # 1.8 billion in September.* (The last column is set aside for now.)

He	PRP	B-NP
reckons	VBZ	B-VP
the	DT	B-NP
current	JJ	I-NP
account	NN	I-NP
deficit	NN	I-NP
will	MD	B-VP
narrow	VB	I-VP
to	TO	B-PP
only	RB	B-NP
#	#	I-NP
1.8	CD	I-NP
billion	CD	I-NP
in	IN	B-PP
September	NNP	B-NP
.	.	O



Part-of-Speech Annotation (Universal Dependencies)

Annotation of: *Do museum labels have an impact on how people look at artworks?*

ID	FORM	LEMMA	UPOS	XPOS	FEATS
1	Do	do	AUX	VBP	Mood=Ind|Number=Plur|Person=3|Tense=Pres|VerbForm=Fin
2	museum	museum	NOUN	NN	Number=Sing
3	labels	label	NOUN	NNS	Number=Plur
4	have	have	VERB	VB	VerbForm=Inf
5	an	a	DET	DT	Definite=Ind|PronType=Art
6	impact	impact	NOUN	NN	Number=Sing
7	on	on	ADP	IN	_
8	how	how	SCONJ	WRB	PronType=Rel
9	people	person	NOUN	NNS	Number=Plur
10	look	look	VERB	VBP	Mood=Ind|Number=Plur|Person=3|Tense=Pres|VerbForm=Fin
11	at	at	ADP	IN	_
12	artworks	artwork	NOUN	NNS	Number=Plur
13	?	?	PUNCT	.	_



Part-of-Speech Annotation (Universal Dependencies)

Annotation of: *Genom skattereformen införs individuell beskattning (särbeskattning) av arbetsinkomster.*

ID	FORM	LEMMA	UPOS	XPOS	FEATS
1	Genom	genom	ADP	PP	_
2	skattereformen	skattereform	NOUN	NN|UTR|SIN|DEF|NOM	Case=Nom|Definite=Def|Gender=Com|Number=Sing
3	införs	införa	VERB	VB|PRS|SFO	Mood=Ind|Tense=Pres|VerbForm=Fin|Voice=Pass
4	individuell	individuell	ADJ	JJ|POS|UTR|SIN|IND|NOM	Case=Nom|Definite=Ind|Degree=Pos|Gender=Com|Number=Sing
5	beskattning	beskattning	NOUN	NN|UTR|SIN|IND|NOM	Case=Nom|Definite=Ind|Gender=Com|Number=Sing
6	(	(	PUNCT	PAD	_
7	särbeskattning	särbeskattning	NOUN	NN|UTR|SIN|IND|NOM	Case=Nom|Definite=Ind|Gender=Com|Number=Sing
8	)	)	PUNCT	PAD	_
9	av	av	ADP	PP	_
10	arbetsinkomster	arbetsinkomst	NOUN	NN|UTR|PLU|IND|NOM	Case=Nom|Definite=Ind|Gender=Com|Number=Plur
11	.	.	PUNCT	MAD	_



Ambiguity

Word	Possible tags	Example of use
that	Subordinating conjunction	*That he can swim is good*
	Determiner	*That white table*
	Adverb	*It is not that easy*
	Pronoun	*That is the table*
	Relative pronoun	*The table that collapsed*
round	Verb	*Round up the usual suspects*
	Preposition	*Turn round the corner*
	Noun	*A big round*
	Adjective	*A round box*
	Adverb	*He went round*
table	Noun	*That white table*
	Verb	*I table that*
might	Noun	*The might of the wind*
	Modal verb	*She might come*
collapse	Noun	*The collapse of the empire*
	Verb	*The empire can collapse*



Part-of-Speech Ambiguity in Swedish

The word *som* in the *Norstedts svenska ordbok* (1999) has three entries:

1. *Om jag vore lika vacker som du, skulle jag vara lycklig.* (conjunction)
2. *Bilen som jag köpte i fjol.* (pronoun)
3. *Som jag har saknat dig.* (adverb)

The part-of-speech distinction can be significant:

Swedish. Compare the pronunciation of *vaken* (adjective) in *Han är aldrig vaken innan klockan sju* with *vaken* (noun) in *Vi fiskade i vaken i sjön*.

English. Compare *object* in *I object to violence* (verb) with *I could see an object* (noun).



Grammatical Features

Main parts of speech	**Features (subcategories)**
Adjective, noun, pronoun	Regular, base, comparative, superlative, interrogative, person, number, case
Adverb	Regular, base, comparative, superlative, interrogative
Article, determiner, preposition	Person, case, number
Verb	Tense, voice, mood, person, number, case



Lexicons: An Excerpt from the Oxford Advanced Learner's Dictionary

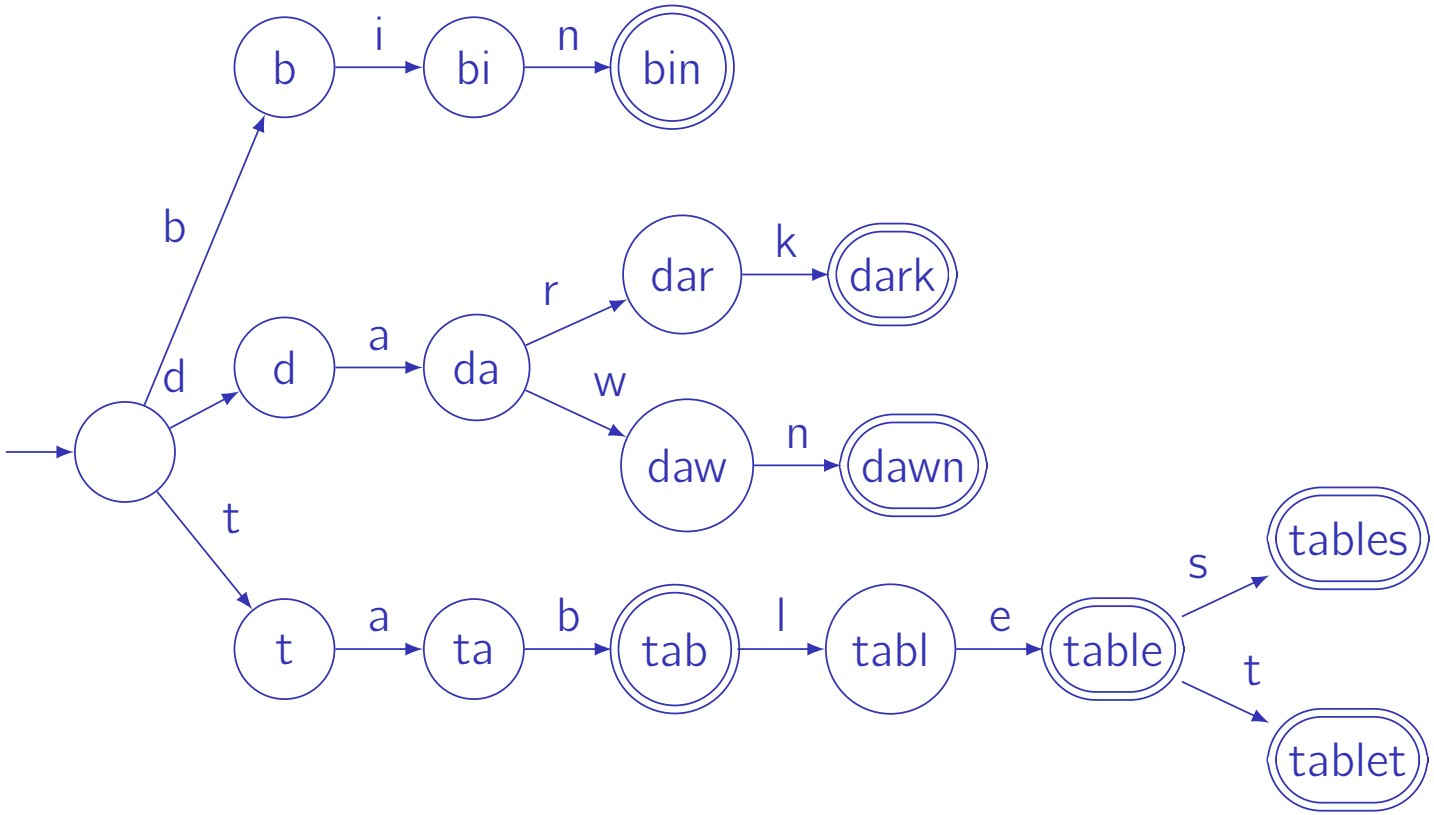
Word	Pronunciation	Syntactic tag	Syllable count or verb pattern
a	@	S-*	1
a	EI	Ki$	1
a fortiori	el ,fOtI'Oral	Pu$	5
a posteriori	el ,p0sterI'Oral	OA$,Pu$	6
a priori	el ,praI'Oral	OA$, Pu$	4
a's	Eiz	Kj$	1
ab initio	&b I'nISI@U	Pu$	5
abaci	'&b@saI	Kj$	3
aback	@'b&k	Pu%	2
abacus	'&b@k@s	K7%	3
abacuses	'&b@k@sIz	Kj%	4
abaft	@'bAft	Pu$,T-$	2
abandon	@'b&nd@n	H0%,L@%	36A,14
abandoned	@'b&nd@nd	Hc%,Hd%,OA%	36A,14
abandoning	@'b&nd@nIN	Hb%	46A,14
abandonment	@'b&nd@nm@nt	L@%	4
abandons	@'b&nd@nz	Ha%	36A,14
abase	@'beIs	H2%	26B
abased	@'beIst	Hc%,Hd%	26B
abasement	@'beIsm@nt	L@%	3

Newer standard:

https://tei-c.org/release/doc/tei-p5-doc/en/html/DI.html



Letter Trees





Morphemes

	Word	Morpheme decomposition
English	*disentangling*	*dis*+*en*+**tangle**+*ing*
	rewritten	*re*+**write**+*en*
French	*désembrouillé*	*dé*+*em*+**brouiller**+*é*
	récrite	*re*+**écrire**+*te*
German	*entwirrend*	*ent*+**wirren**+*end*
	wiedergeschrieben	*wieder*+*ge*+**schreiben**+*en*



Inflection

	Plural of nouns	**Morpheme decomposition**
English	*hedgehogs*	*hedgehog+s*
	churches	*church+es*
	sheep	*sheep+∅*
French	*hérissons*	*hérisson+s*
	chevaux	*cheval+ux*
German	*Gründe*	*Grund+(¨)e*
	Hände	*Hand+(¨)e*
	Igel	*Igel+∅*



Derivation

Creation of a new word:

	English	**French**	**German**
Prefixes	**fore***see*, **un***pleasant*	**pré***voir*, **dé***plaisant*	**vorher***sehen*, **un***angenehm*
Suffixes	*manage***able**, *rigor***ous**	*gér***able**, *rigour***eux**	*vorsicht***ich**, *streit***bar**



Morphological Processing

Generation →

English		**French**		**German**	
dog+s	*dogs*	*chien+s*	*chiens*	*Hund+e*	*Hunde*
work+ing	*working*	*travailler+ant*	*travaillant*	*arbeiten+end*	*arbeitend*
un+do	*undo*	*dé+faire*	*défaire*		

← Parsing



Language Differences (Source: Xerox)

Language	# stems	# inflected forms		Lex. size (kb)
English	55,000	240,000		200–300
French	50,000	5,700,000		200–300
German	50,000	350,000	or	450
		infinite	(compounding)	
Japanese	130,000	200	suffixes	500
		20,000,000	word forms	500
Spanish	40,000	3,000,000		200–300



Two-Level Morphology

Modern morphological parsers are based on the two-level model of Kimmo Koskenniemi (1983).
The model relates the surface form of a word (the form that appears in a text) to its lexical (or underlying) form (its sequence of morphemes):

Surface:	disentangled
Lexical (or underlying):	dis+en+tangle+ed



Examples

Generation: Lexical to surface form →		
English	*dis+en+tangle+ed*	*disentangled*
	happy+er	*happier*
	move+ed	*moved*
French	*dés+em+brouiller+é*	*désembrouillé*
	dé+chanter+erons	*déchanterons*
German	*ent+wirren+end*	*entwirrend*
	wieder+ge+schreiben+en	*wiedergeschrieben*
Parsing: ← Surface to lexical form		



Aligning the Two Forms

English	dis+en+tangle+ed	happy+er	move+ed
	⇅ ...	⇅ ...	⇅ ...
	dis0en0tangl00ed	happi0er	mov00ed
French	dé+chanter+erons	cheval+ux	cheviller+é
	⇅ ...	⇅ ...	⇅ ...
	dé0chant000erons	cheva00ux	chevill000é
German	singen+st	Grund+¨e	Igel+∅
	⇅ ...	⇅ ...	⇅ ...
	singe00st	Gründ00e	Igel00



Interpreting the Morphemes

Suffixes carry grammatical information: the French ending *erons*, for example, encodes verb + future + first person + plural.
Morphological parsers can therefore represent the lexical form as a concatenation of the stem and its grammatical features rather than the stem and the surface suffix.
The Xerox parser produces the following analyses for *disentangled* and *happier*:

```
disentangle+Verb+PastBoth+123SP
happy+Adj+Comp
```

where `+Verb` marks a verb, `+PastBoth` indicates either past tense or past participle, and `+123SP` stands for any person, singular or plural; `+Adj` marks an adjective and `+Comp` a comparative.



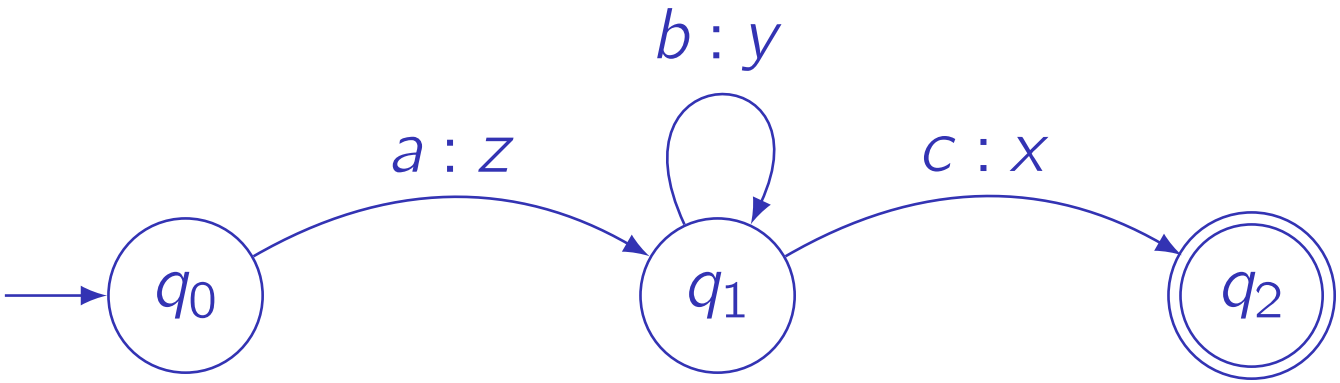
Aligning Morphemes and Features

Lexical:	d	i	s	e	n	t	a	n	g	l	e	+Verb	+PastBoth	+123sp
Surface:	d	i	s	e	n	t	a	n	g	l	0	0	e	d

Lexical:	h	a	p	p	y	+Adj	+Comp
Surface:	h	a	p	p	i	e	r



Transducers



The string abbbc is transduced into zyyyx.



Mathematical Definition of a Finite-State Transducer

1. Q is a finite set of states.
2. Σ is a finite set of symbol pairs $i : o$, where i belongs to the input alphabet and o to the output alphabet. Both alphabets may include the empty symbol ε.
3. q_0 is the start state ($q_0 \in Q$).
4. F is the set of final states ($F \subseteq Q$).
5. δ is the transition function $Q \times \Sigma \rightarrow Q$, where $\delta(q, i : o)$ returns the state reached from q on the input–output pair $i : o$.

For the example transducer we have $Q = \{q_0, q_1, q_2\}$,
$\Sigma = \{a : z, b : y, c : x\}$,
$\delta = \{\delta(q_0, a : z) = q_1, \delta(q_1, b : y) = q_1, \delta(q_1, c : x) = q_2\}$, and $F = \{q_2\}$.



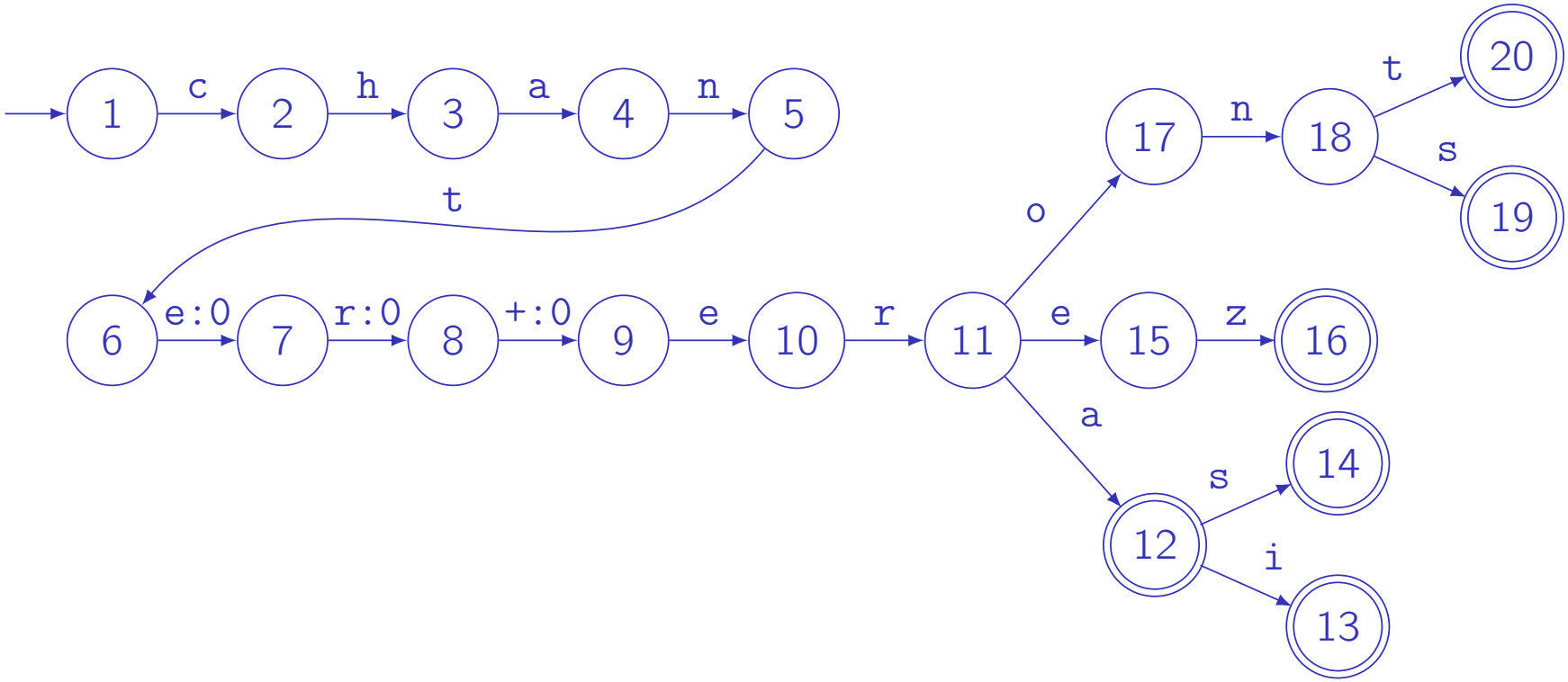
French Verb Transducers for *chanter*

Number\Person	First	Second	Third
singular	*chanterai*	*chanteras*	*chantera*
plural	*chanterons*	*chanterez*	*chanteront*

Number\Pers.	First	Second	Third
singular	`chanter+erai`	`chanter+eras`	`chanter+era`
	`chant000erai`	`chant000eras`	`chant000era`
plural	`chanter+erons`	`chanter+erez`	`chanter+eront`
	`chant000erons`	`chant000erez`	`chant000eront`

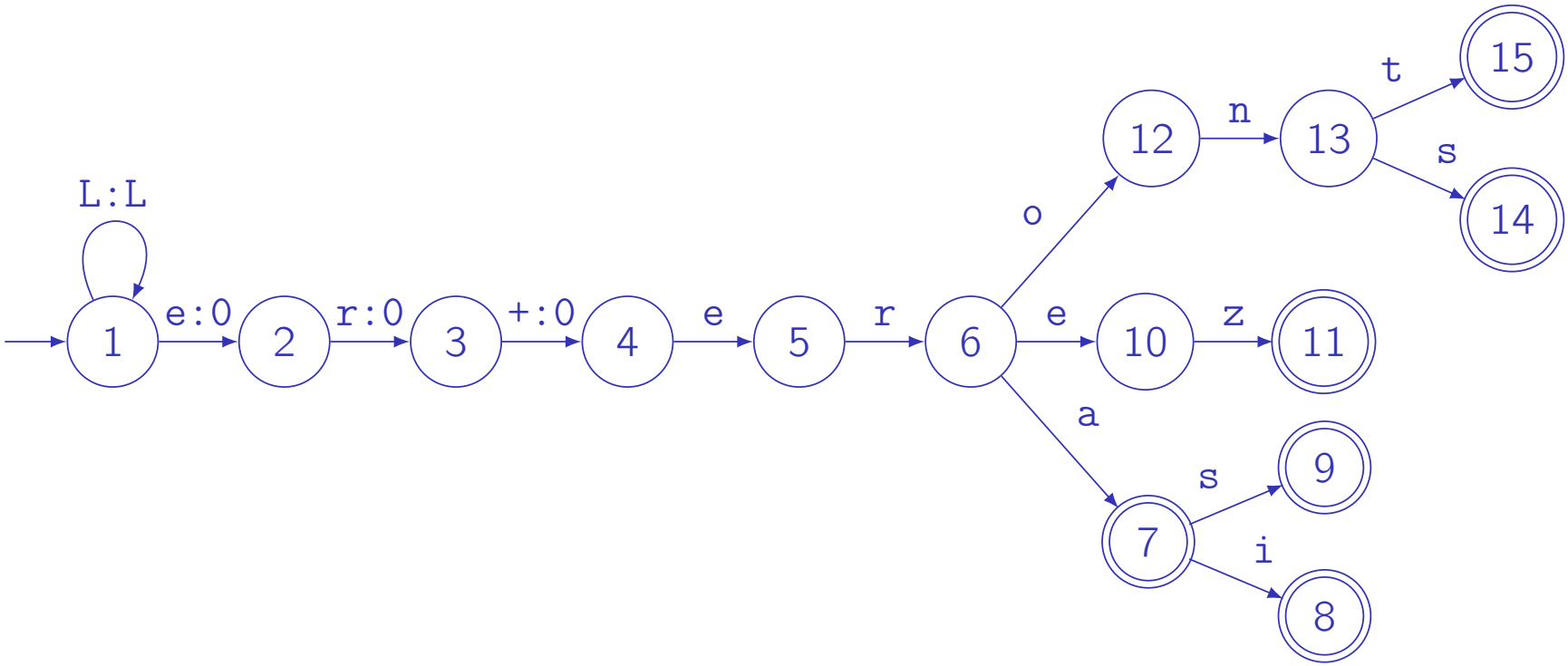


Transducer for *chanter*





French Verb Transducers: Future, First Group



Such transducers can be implemented in Prolog or with OpenFst (https://www.openfst.org/).



Romance Languages

Language	Number\Person	First	Second	Third
Italian				
	singular	*canterò*	*canterai*	*canterà*
	plural	*canteremo*	*canterete*	*canteranno*
Spanish				
	singular	*cantaré*	*cantarás*	*cantará*
	plural	*cantaremos*	*cantaréis*	*cantarán*
Portuguese				
	singular	*cantarei*	*cantarás*	*cantará*
	plural	*cantaremos*	*cantareis*	*cantarão*



Course Content: Optional Part

Starting in 2023, the remaining slides in this document are no longer part of the course.

You may nevertheless read them if you are curious.



Ambiguity

In the future-tense transducer there is no ambiguity: each surface form corresponds to a unique lexical form and a unique final state.
This is not the case for the present tense:

(je) chante 'I sing'
(il) chante 'he sings'

Number\Person	**First**	**Second**	**Third**
singular	*chante*	*chantes*	*chante*
plural	*chantons*	*chantez*	*chantent*

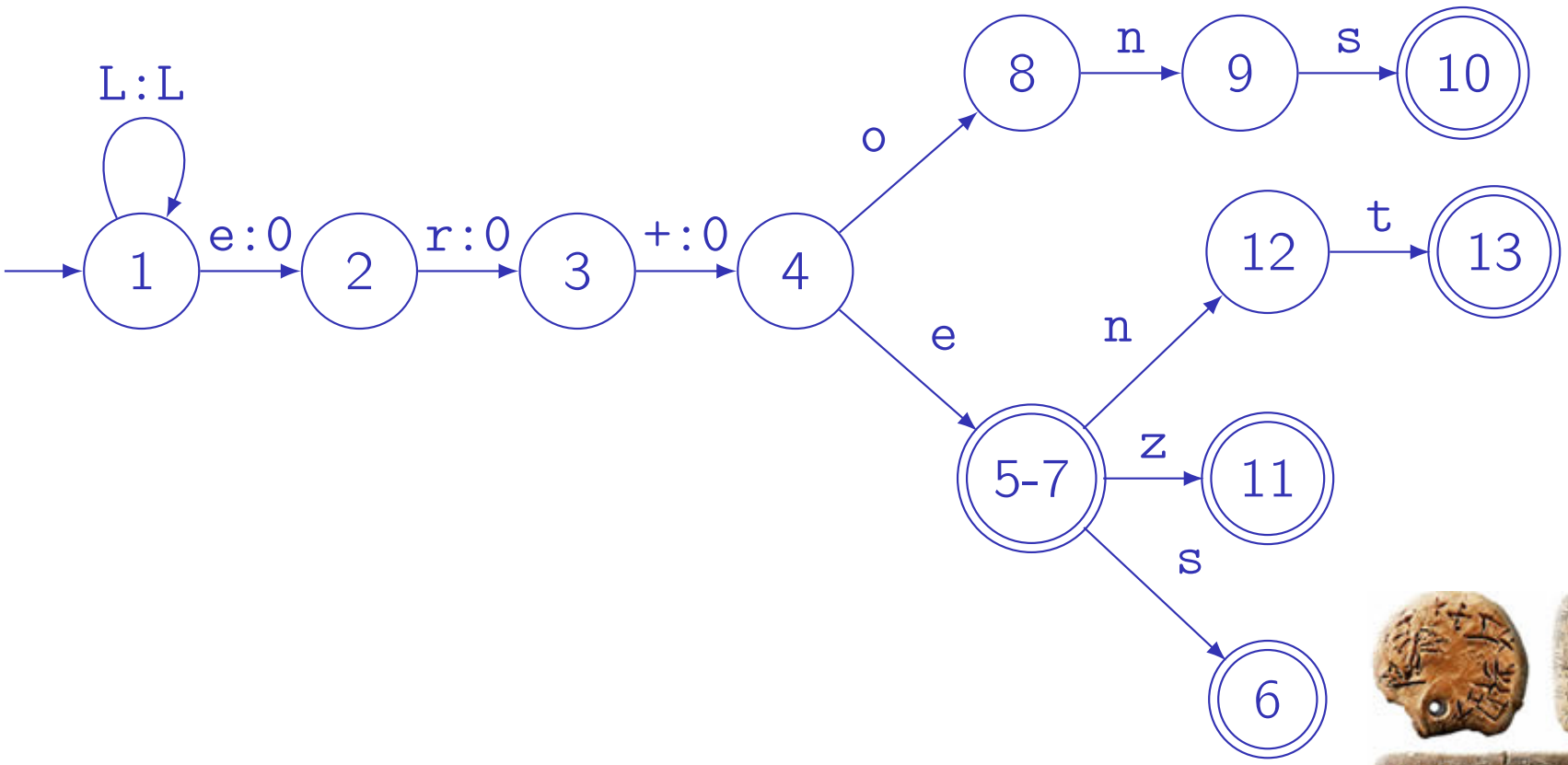


Transducer Ambiguity

Final states 5 and 7 coincide.

A Prolog implementation is analogous to that of the future tense.

By backtracking, the transducer can return all final states and thereby expose the morphological ambiguity.





Koskenniemi's Rules

Koskenniemi described morphology with declarative rules that refer to left and right contexts and employ the operators $\Rightarrow$, $\Leftarrow$, $\Leftrightarrow$, or $/\Leftarrow$.
In English a lexical *y* may correspond to a surface *i*, as in *happier*.
This happens when *y* is preceded by a consonant and followed by *-er*, *-ed* or *-s*:

1. `y:i` $\Leftarrow$ `C:C __ +:0 e:e r:r`
2. `y:i` $\Leftarrow$ `C:C __ +:e s:s`
3. `y:i` $\Leftarrow$ `C:C __ +:0 e:e d:d`



Two-Level Rules

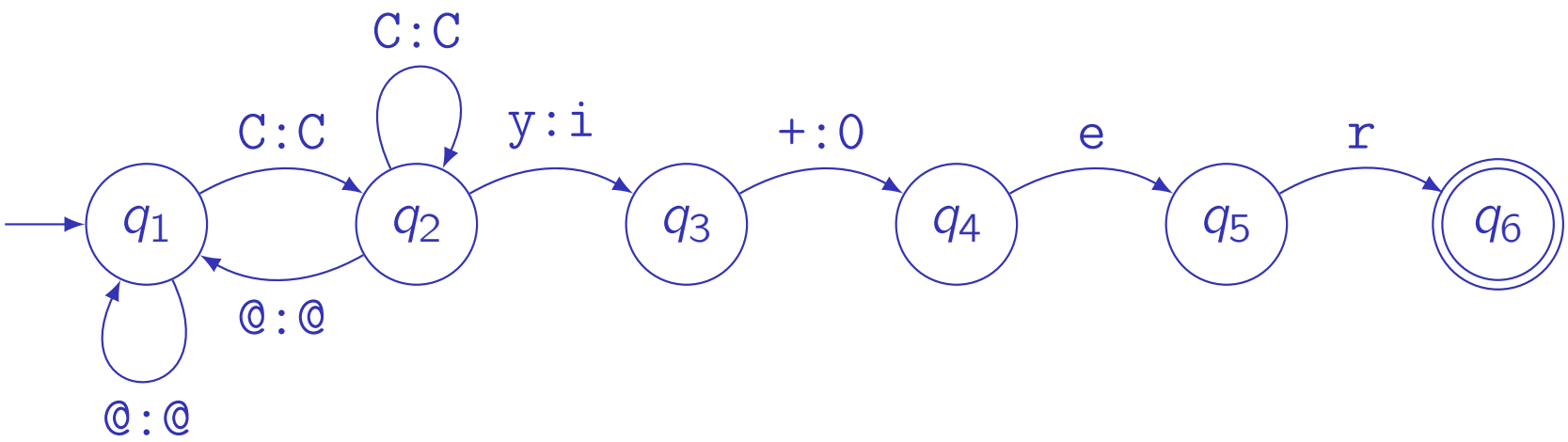
Lexical-to-surface transduction is expressed by rules of the following kinds:

Rules			Description
`a:b`	$\Rightarrow$	`lc __ rc`	`a` is realized as `b` **only** in the context `lc __ rc`
`a:b`	$\Leftarrow$	`lc __ rc`	`a` is **always** realized as `b` in the context `lc __ rc`
`a:b`	$\Leftrightarrow$	`lc __ rc`	`a` is realized as `b` **always and only** in the context `lc __ rc`
`a:b`	$/\Leftarrow$	`lc __ rc`	`a` is **never** realized as `b` in the context `lc __ rc`



Parallel Rules

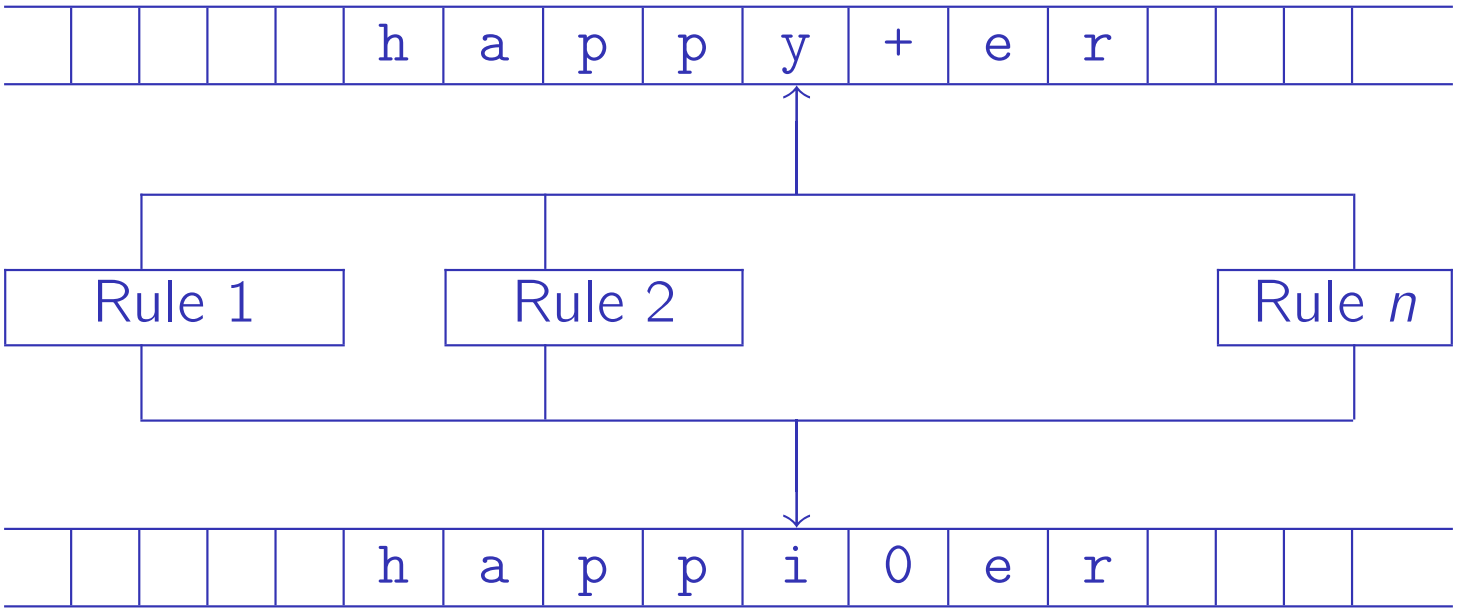
All applicable rules are applied in parallel (provided their contexts match).





Rules and Transducers

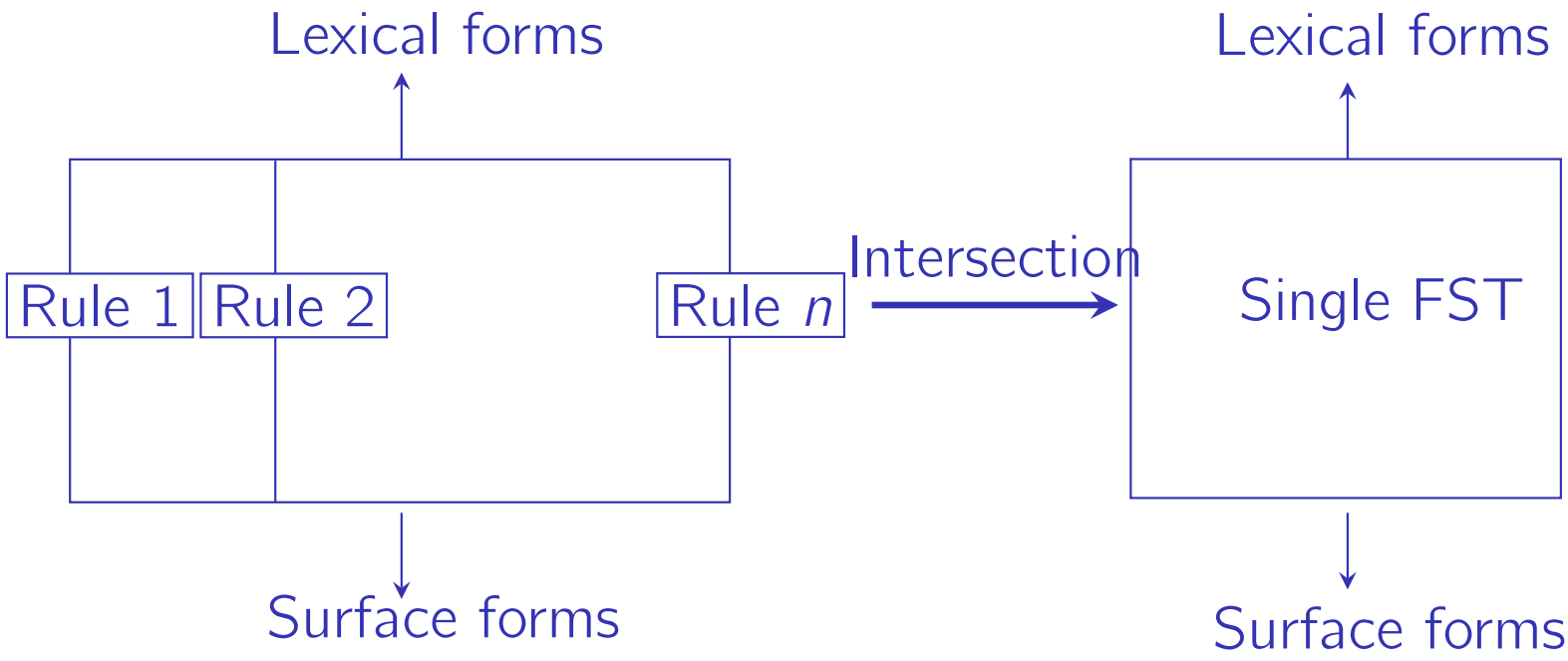
Each rule can be compiled into an equivalent finite-state transducer.





Rule Intersection

The parallel transducers are subsequently combined into a single transducer by intersection.





Problems with Intersection

The intersection of two finite-state automata is always a finite-state automaton.
This is not always true for finite-state transducers.
Kaplan and Kay (1994) showed that when the surface and lexical strings have the same length (no ε-transitions), the intersection is itself a transducer.
This property is sufficient for practical rule intersection.
In practice, transducers derived from two-level rules are intersected by treating the ε symbol as an ordinary symbol (Beesley & Karttunen 2003, p. 55).



Xerox

Originally the rules were compiled by hand.
This quickly becomes intractable when rules conflict or when rule contexts interact with the symbols being transduced.
The problem is solved by a compiler that automatically constructs transducers from two-level rules.
Xerox's XFST is a publicly available tool and, to date, the only robust implementation of a morphological rule compiler.



Morphology of French Verbs

We have so far used a stem together with a set of suffixes for regular French verbs.
Irregular French verbs are considerably more complex.
Chanod (1994) provides an example of their decomposition into simple rules.

Infinitive	**courir**	**dormir**	**battre**	**peindre**	**écrire**
1st sg.	cour**s**	**dors**	**bats**	**peins**	écris
2nd sg.	cour**s**	**dors**	**bats**	**peins**	écris
3rd sg.	cour**t**	**dort**	**bat**	**peint**	écrit
1st pl.	cour**ons**	dormons	battons	peignons	**écrivons**
2nd pl.	cour**ez**	dormez	battez	peignez	**écrivez**
3rd pl.	cour**ent**	dorment	battent	peignent	**écrivent**



French Morphology

Lexical form: stem	dormir	+IndP	+SG +P1
	↕		↕
Intermediate form: inflection	dorm	+IndP	+SG +P1
	↕		↕
Intermediate form: deletion of *m* before *s*	dorm		s
	↕		↕
Surface form:	dor		s

From *peindre* to *peins*:

$$\texttt{n:0} \Leftrightarrow \texttt{g __ [s|t]}$$



Composition and Intersection

